Credit Card Fraud Detection & Anomaly Audit Report

Production Intelligence Report | Generated: October 01, 2026 - 05:58:52 UTC | Strict Compliance

1. Executive Transaction Summary

Total Scored	**1,202**	**Flagged / Anomalous**	**80**
Anomaly Rate	**3.00%**	**Avg Transaction**	**$106.84**
Active Model	**Isolation Forest**	**Model Version**	**v1.0.0**

2. Active Model Performance & Benchmark Metrics

Metric	Score	Benchmark Guideline	Operational Status
Precision	1.0000	Minimizes false customer card declines	**Optimal**
Recall (Sensitivity)	0.8833	Maximizes fraud dollar capture	**Optimal**
F1-Score	0.9381	Harmonic balance of Precision & Recall	**Robust**
PR-AUC (Avg Precision)	0.9905	Primary metric for extreme imbalance	**High**
ROC-AUC	0.9996	Overall ranking capability across thresholds	**Superior**
False Positive Rate (FPR)	0.0000	Under 2% friction on genuine cardholders	**Within SLA**

3. Continuous Learning & Data Drift Diagnostics

Diagnostic Check	Observed Value	Threshold SLA	Automated Action
Population Stability (PSI)	0.0520	< 0.25 (Standard Threshold)	**STABLE**
Retraining Mode	Batch & Incremental Trigger	Continuous Drift Sentinel	**Enabled**
Last Retraining Event	Baseline Deployment	Scheduled / Drift Triggered	**SUCCESS**

4. Sample Flagged Transactions & Root-Cause Explainability

TX ID	Amount	Merchant / Channel	Score	Classification	Contributing Risk Factors
TX-1790813052 .789568	$28.80	electronics (Card)	**0.496**	**Suspicious**	["Unusual multidimensional latent behavioral
TX-1790813052 .78916	$991.68	electronics (Card)	**0.820**	**Potentially Fraudulent**	["Unusual transaction time (00:00 hours)", "A
TX-1790813052 .788815	$55.81	supermarket (Card)	**0.515**	**Suspicious**	["Unusual transaction time (04:00 hours)", "H
TX-1790813052 .788719	$104.14	crypto_exchange (Card)	**0.483**	**Suspicious**	["Elevated merchant category risk (Crypto Exc
TX-1790813052 .78867	$11.17	electronics (Card)	**0.849**	**Potentially Fraudulent**	["Unusual transaction time (00:00 hours)", "E

TX-1790813052 .788525	$124.56	pharmacy (Card)	**0.610**	**Suspicious**	["Unusual transaction time (04:00 hours)", "T
TX-1790813052 .78838	$29.28	grocery (Card)	**0.483**	**Suspicious**	["Unusual multidimensional latent behavioral
TX-1790813052 .788331	$32.04	pharmacy (Card)	**0.504**	**Suspicious**	["Unusual multidimensional latent behavioral
TX-1790813052 .787655	$14.63	pharmacy (Card)	**0.509**	**Suspicious**	["Unusual transaction time (01:00 hours)", "H
TX-1790813052 .786875	$326.41	supermarket (Card)	**0.487**	**Suspicious**	["Unusual multidimensional latent behavioral